Physiology and Pharmacology

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Physiology and Pharmacology (A13844)

Short Title:	Physiology and Pharmacology		
Faculty	Science and Computing		
Department	Science		
Cluster	Biology		
Author	EMCNEELA		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module aims to provide students with an overview of human anatomy and physiology of the main body systems, and the role of pharmacological interventions. The module also provides the students with an introduction to the pharmacological actions of drugs, including pharmacodynamics and pharmacokinetics. In particular, the course focuses on the pharmacological properties and in vivo behaviour of biopharmaceuticals including bioactive proteins, peptides, nucleic acids and monoclonal antibodies. Pharmacological principles of these biomacromolecules and challenges to their pharmacological action in vivo are discussed and illustrated with examples from the literature.

Programmes

PHAR-0018	BSc (Hons) in Applied Biology with Quality Management (WD_SQMBI_B)	4 / 2 / M
PHAR-0018	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 8 / M

Indicative Content

- Organisation of cells, tissues and organs and relationships to functions
- An introduction to the major systems in the body: cardiovascular, nervous, endocrine, gastrointestinal and renal
- Introduction to pharmacological concepts; drugs and how they act; pharmacology in the 21st century; introduction to pharmacodynamics and pharmacokinetics
- Identification of drug targets and drug design; bioassays, animal models and clinical trials
- Pharmacodynamics and mechanisms of action of selected biopharmaceuticals; receptors and signalling
- Pharmacokinetics: absorption, distribution, metabolism, elimination, excretion, bioavailability and half-life
- Pharmacokinetics of some selected biopharmaceuticals including proteins and peptides, monoclonal antibodies and nucleic-acid based biologics
- Drug-drug interactions; immunogenicity of biologics
- Individual variation in drug responsiveness and pharmacogenomics

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Comment on how the structure of cells, tissues and organs is related to their physiological functions.
2. Interpret the theories and concepts underlying pharmacodynamics and pharmacokinetics.
3. Evaluate the importance of bioassays, animal models and clinical trials in the development of biologics for clinical use.
4. Distinguish between small molecule drugs and biologics and determine the relationship between the properties of biologics and their unique pharmacokinetic characteristics in vivo.
5. Select new and emerging biopharmaceuticals and debate their pharmacological strengths and weaknesses.
6. Present findings on research into the pharmacological properties of novel pharmaceutical drugs.

Learning and Teaching Methods

- This module will use a blended learning approach combining face-to-face lectures with e-learning available through the platform Moodle.
- Student engagement will be enhanced through use of the interactive e-learning platform, in-class discussions through workshops, research of current relevant literature and informative presentations.
- A learning and teaching strategy appropriate to level 8 will be adopted throughout with opportunities for formative assessment to encourage reflective, critical thinking.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	1,2,5,6
Assessment	10%	1,2,5,6'

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of physiology and pharmacology. Unable to perform critical thinking and analysis.

40%-49%: Shows a limited understanding of the fundamentals of physiology and pharmacology. Has difficulty with critical thinking and analysis.

50%-59%: Shows a reasonable understanding of the fundamentals of physiology and pharmacology. Shows some ability to analyse and think critically and independently.

60%-69%: Shows a good understanding of the fundamentals of physiology and pharmacology. Has good critical thinking and analyses and debates in an effective manner.

70%-100%: All of the above to an excellent level. Demonstrates a highly developed and extensive comprehension of the subject manner. Shows ability to evaluate, debate and relate topics and shows evidence of extensive independent learning and knowledge of current literature and recent developments in the area.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Independent Learning	99	

Essential Material

- Meibohm, B. _Pharmacokinetics and Pharmacodynamics of Biotech Drugs: Principles and Case Studies in Drug Development_. United States: Wiley-Blackwell, 2006.

Supplementary Material

- "Drug Development Technology." http://www.drugdevelopment-technology.com
- "PubMed Search Engine." http://www.ncbi.nlm.nih.gov/pubmed/
- Crommelin, D.J.A., R.D. Sindelar and B. Meibohm. _Pharmaceutical Biotechnology: Fundamentals and Applications_. 4th ed. New York: Springer-Verlag, 2013.
- Hollinger, M.A. _Introduction to Pharmacology_. 2nd ed. London; New York: Taylor and Francis, 2003.
- Wilson, K.J.W. _Ross and Wilson Anatomy and Physiology in Health and Illness_. 11th ed. Edinburgh: Churchill Livingstone, 2010.

Requested Resources

- Lecture Room: Loose Seated